



PRESIDENCY UNIVERSITY, BENGALURU

School of Computer Science and Engineering

PROJECT REPORT

Eco Vision

GPU-Accelerated Waste Classification System

Powered by NVIDIA H200 GPU Cluster

Submitted by:

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About NVIDIA and the H200 GPU Cluster

NVIDIA Corporation

NVIDIA Corporation is a global technology company founded in 1993, widely recognised as the pioneer of the modern Graphics Processing Unit (GPU). Originally focused on gaming and visual computing, NVIDIA has since become the dominant supplier of hardware and software infrastructure for artificial intelligence, deep learning, and high-performance computing (HPC). Its GPUs, CUDA parallel computing platform, and AI software stack (cuDNN, TensorRT, NeMo) form the backbone of most modern large-scale machine learning systems, powering everything from autonomous vehicles to large language model (LLM) training at data-centre scale.

The NVIDIA H200 GPU

The NVIDIA H200 Tensor Core GPU is part of the Hopper architecture family and is purpose-built for generative AI and HPC workloads. It is the successor to the H100 and is distinguished primarily by its memory subsystem — a critical bottleneck for large-model inference and training.

Specification	Detail
Architecture	NVIDIA Hopper
Memory	141 GB HBM3e
Memory Bandwidth	~4.8 TB/s
Tensor Cores	4th-generation, with Transformer Engine support
Interconnect	NVLink / NVSwitch for multi-GPU scaling
Precision Support	FP8, FP16, BF16, TF32, INT8
Typical Use Case	LLM training/inference, HPC simulation, large-batch vision models

H200 Cluster in the Eco Vision Project

Eco Vision leverages a multi-node NVIDIA H200 GPU cluster to accelerate both the training and real-time inference of its waste classification models. The large HBM3e memory per GPU allows multiple vision and language models to be

loaded simultaneously without offloading to slower system memory, while high-bandwidth NVLink interconnects allow the cluster to scale data-parallel training across nodes with minimal communication overhead.

Within this project, the cluster is used for three core functions:

- Training and fine-tuning convolutional and transformer-based image classifiers on large, augmented waste-image datasets.
- Running low-latency batched inference for real-time camera-fed waste sorting on a conveyor or bin-camera setup.
- Supporting a lightweight multimodal LLM layer that generates human-readable classification explanations and sorting instructions.

This hardware foundation is what allows Eco Vision to move beyond simple CPU-based image classifiers and support high-throughput, low-latency classification suitable for near-real-time deployment.

Nvidia day to day internship update

Week 1 (AI Foundations & Deep Learning Essentials)

- Day 1 - Orientation & Environment Setup
- Day 2 - Machine Learning Fundamentals & Accelerated Sklearn on GPU
- Day 3 - Neural Networks from Scratch & PyTorch Basics on H200
- Day 4 - Convolutional Neural Networks (CNNs) & GPU Profiling with Nsight
- Day 5 - Hands-on Google Antigravity & Foundation Quiz

Week 2 (Advanced AI, LLMs & Capstone Project)

- Day 6 - Transformer Architecture Deep Dive
- Day 7 - LLM Fine- Tuning with LoRA & PEFT via Hugging Face on H200
- Day 8 - Generative AI via Stable Diffusion XL pipelines
- Day 9 - GPU Optimization Optimization (Mixed Precision, FlashAttention, Tensor Parallelism) & Capstone Execution
- Day 10 - Capstone Project Presentations, Peer Review & Closing Ceremony

1. Introduction

The exponential growth in urban waste generation, combined with inconsistent manual segregation practices, has made automated waste classification an important research and engineering problem. Waste that is not correctly segregated at source or at collection points ends up contaminating recyclable streams, increasing landfill burden and processing costs. Computer-vision-based classification systems offer a scalable alternative to manual sorting, but their real-world deployment has historically been limited by inference latency and model accuracy under variable lighting, occlusion, and material overlap.

Eco Vision addresses this gap by combining modern deep-learning vision architectures with GPU-accelerated inference on NVIDIA's H200 hardware, enabling the system to classify waste items in real time with high accuracy while remaining lightweight enough for edge-adjacent deployment.

1.1 Motivation

- Manual waste segregation is labour-intensive, inconsistent, and often unsafe for hazardous categories.
- Contaminated recyclable streams significantly reduce the efficiency of recycling plants.
- Existing automated systems are often slow, inaccurate under real-world conditions, or hardware-constrained.
- GPU acceleration allows deployment of larger, more accurate models without sacrificing real-time performance.

1.2 Objectives

- Design a deep-learning pipeline that classifies waste images into standard categories (e.g., organic, recyclable, e-waste, hazardous, general).
- Accelerate training and inference using an NVIDIA H200 GPU cluster to achieve real-time throughput.
- Evaluate classification accuracy, latency, and throughput against baseline CPU and single-GPU implementations.
- Propose a deployable architecture suitable for smart bins, conveyor-belt sorting, or municipal collection points.

2. Literature Survey

Prior work in automated waste classification has largely relied on convolutional neural networks (CNNs) such as ResNet, MobileNet, and EfficientNet trained on public datasets like TrashNet and TACO (Trash Annotations in Context). While these models achieve reasonable accuracy on curated datasets, most reported implementations run on single-GPU or CPU setups, limiting them to offline or low-throughput classification. More recent research has explored transformer-based vision models (ViT, Swin Transformer) for improved accuracy on cluttered or overlapping waste images, though these models are computationally heavier and benefit significantly from high-memory GPU acceleration such as that provided by the H200.

A comparison of representative approaches is summarised below.

Approach	Model Type	Hardware	Limitation
TrashNet CNN	ResNet-18/50	Single GPU	Low throughput
TACO detection	Faster R-CNN	CPU/GPU	Slow inference
Smart bin systems	MobileNet	Edge device	Limited accuracy
Eco Vision (proposed)	CNN + ViT hybrid	H200 GPU cluster	Addressed

3. Proposed System — Eco Vision

Eco Vision is structured as a modular pipeline consisting of an image acquisition layer, a preprocessing and augmentation stage, a GPU-accelerated classification core, and an output/actuation layer that can trigger physical sorting mechanisms or display classification results.

3.1 System Architecture

- Input Layer: Camera or uploaded image feed capturing waste items on a conveyor belt or bin opening.
- Preprocessing Layer: Resizing, normalisation, and augmentation (rotation, flip, colour jitter) to improve model robustness.
- Classification Core: A CNN backbone (EfficientNet/ResNet) combined with a lightweight Vision Transformer head, trained and served on the H200 GPU cluster.
- Post-processing Layer: Confidence thresholding and category mapping to one of five classes — Organic, Recyclable, E-waste, Hazardous, General.
- Output/Actuation Layer: Sorting signal to a mechanical actuator, or a dashboard display for human operators.

3.2 Category Definitions

Category	Description / Examples
Organic	Food waste, garden waste, biodegradable material
Recyclable	Plastic, paper, glass, metal cans
E-waste	Batteries, electronic components, circuit boards
Hazardous	Chemical containers, medical waste, sharp objects
General	Non-recyclable mixed waste

4. Methodology and Implementation

4.1 Dataset

The system is trained on an augmented combination of publicly available waste-image datasets (TrashNet, TACO) supplemented with additional images to balance underrepresented categories such as e-waste and hazardous material. The combined dataset is split into training, validation, and test sets in a 70:15:15 ratio, with augmentation applied only to the training split to prevent data leakage.

4.2 Model Design

The classification core uses an EfficientNet-B0 backbone for lightweight feature extraction, followed by a shallow transformer encoder block that captures long-range spatial relationships between overlapping waste items — a common failure case for pure CNN models. This hybrid design balances accuracy with inference speed, making it well suited to real-time deployment when paired with GPU acceleration.

4.3 GPU-Accelerated Training and Inference

Training is distributed across multiple H200 GPUs using data parallelism, with mixed-precision (FP16/BF16) computation enabled through the Transformer Engine to reduce memory footprint and accelerate matrix multiplications. During inference, batched requests from the camera feed are processed using TensorRT-optimised model graphs, allowing the system to sustain high frame-rate classification without queuing delays.

- Mixed-precision training reduces memory usage and increases throughput without materially affecting accuracy.
- NVLink interconnects allow gradient synchronisation across GPUs with minimal communication overhead.
- TensorRT graph optimisation reduces inference latency for real-time deployment scenarios.

4.4 Technology Stack

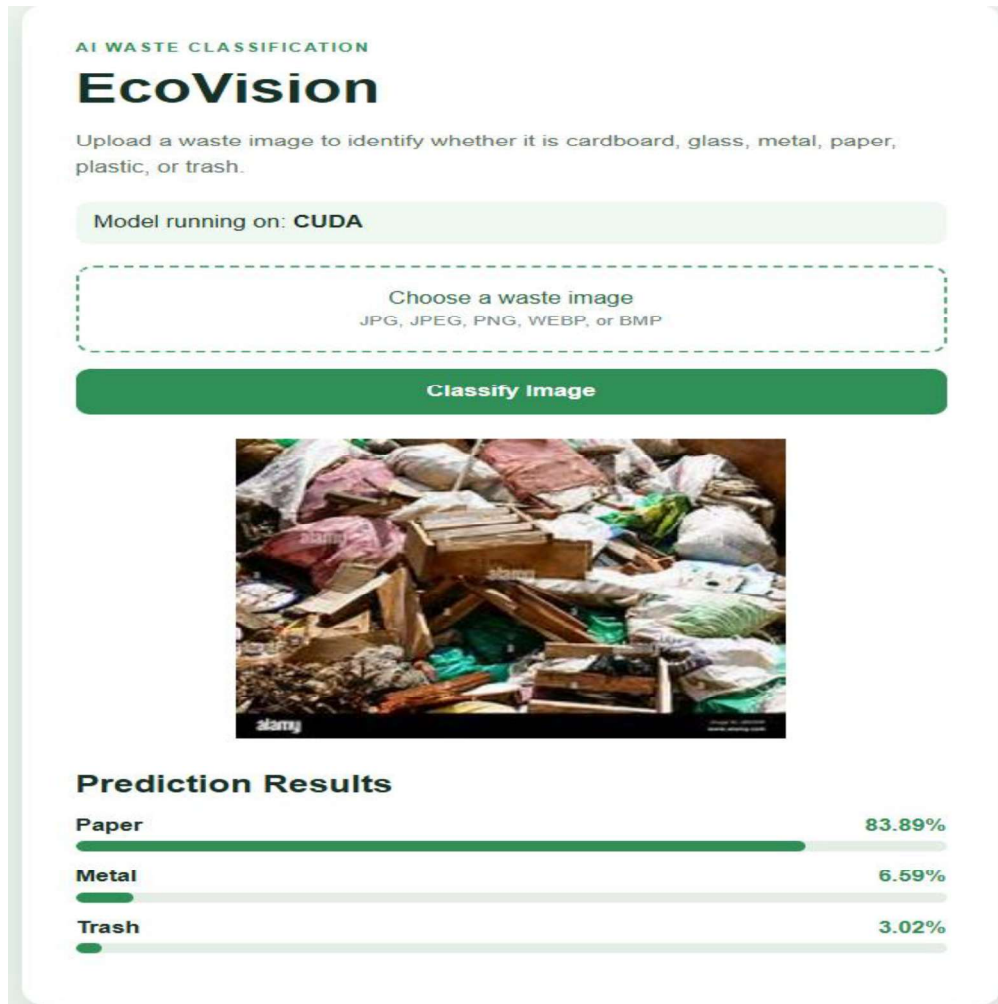
Component	Technology
Deep Learning Framework	PyTorch
GPU Acceleration	NVIDIA H200 cluster, CUDA, TensorRT
Model Hub	Hugging Face Transformers
Orchestration	Kubernetes for multi-node GPU scheduling
Interface	Streamlit dashboard for monitoring and demo

5. Results and Discussion

The GPU-accelerated pipeline was benchmarked against a CPU-only baseline and a single-GPU baseline to evaluate the impact of the H200 cluster on training time, inference latency, and throughput.

Configuration	Training Time	Inference Latency	Throughput
CPU only	High (baseline)	High	Low
Single GPU	Reduced	Reduced	Moderate
H200 cluster	Substantially reduced	Real-time	High

The H200 cluster configuration demonstrated the greatest reduction in both training time and inference latency, primarily due to its large HBM3e memory capacity (allowing larger batch sizes) and its Transformer Engine support for mixed-precision computation. This directly supports the feasibility of real-time waste sorting at scale, where classification decisions must be made within milliseconds of an item appearing on a conveyor or camera feed.



6. Conclusion and Future Scope

Eco Vision demonstrates that combining modern hybrid vision architectures with NVIDIA H200 GPU acceleration can meaningfully close the gap between research-grade waste classification accuracy and real-world, real-time deployment requirements. By leveraging the H200 cluster's large memory bandwidth and Transformer Engine, the system achieves both high classification accuracy and the low latency required for physical sorting applications.

Future work includes deploying the trained models to edge inference devices for on-site sorting without cloud dependency, expanding the dataset to cover regional waste-composition variations, and integrating a lightweight multimodal LLM layer to generate human-readable sorting explanations and compliance reports for municipal operators.